EXAMPLE 2.6

The resistance of a wire is 20 Ω. What will be new resistance, if it is stretched uniformly 8 times its original length?

Solution

$R_1 = 20\ \Omega$, $R_2 = ?$

Let the original length of the wire (l_1) be l.

New length, $l_2 = 8l_1$ *(i.,e)* $l_2 = 8l$

Original resistance, $R_1 = \rho\dfrac{l_1}{A_1}$

New resistance $R_2 = \rho\dfrac{l_2}{A_2} = \dfrac{\rho(8l)}{A_2}$

Though the wire is stretched, its volume remains unchanged.

Initial volume = Final volume

$A_1 l_1 = A_2 l_2$, $\quad A_1 l = A_2(8l)$

$$\frac{A_1}{A_2} = \frac{8l}{l} = 8$$

By dividing equation for R_2 by equation for R_1, we get

$$\frac{R_2}{R_1} = \frac{\rho(8l)}{A_2} \times \frac{A_1}{\rho l}$$

$$\frac{R_2}{R_1} = \frac{A_1}{A_2} \times 8$$

Substituting the value of $\dfrac{A_1}{A_2}$, we get

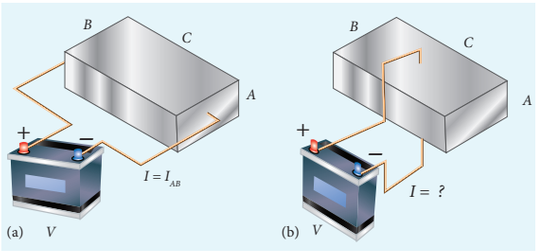
$$\frac{R_2}{R_1} = 8 \times 8 = 64$$

$$R_2 = 64 \times 20 = 1280\ \Omega$$

Hence, stretching the length of the wire has increased its resistance.

EXAMPLE 2.7

Consider a rectangular block of metal of height A, width B and length C as shown in the figure.



If a potential difference of V is applied between the two faces A and B of the block (figure (a)), the current I_{AB} is observed. Find the current that flows if the same potential difference V is applied between the two faces B and C of the block (figure (b)). Give your answers in terms of I_{AB}.

Solution

In the first case, the resistance of the block

$$R_{AB} = \rho\frac{\text{length}}{\text{Area}} = \rho\frac{C}{AB}$$

The current $I_{AB} = \dfrac{V}{R_{AB}} = \dfrac{V}{\rho}\cdot\dfrac{AB}{C}$ (1)

In the second case, the resistance of the block $R_{BC} = \rho\dfrac{A}{BC}$

The current $I_{BC} = \dfrac{V}{R_{BC}} = \dfrac{V}{\rho}\cdot\dfrac{BC}{A}$ (2)

To express I_{BC} interms of I_{AB}, we multiply and divide equation (2) by AC, we get

$$I_{BC} = \frac{V}{\rho}\cdot\frac{BC}{A}\frac{AC}{AC} = \left(\frac{V}{\rho}\cdot\frac{AB}{C}\right)\cdot\frac{C^2}{A^2} = \frac{C^2}{A^2}\cdot I_{AB}$$

Since $C > A$, the current $I_{BC} > I_{AB}$



The human body contains a large amount of water which has low resistance of around 200 Ω and the dry skin has high resistance of around 500 k Ω. But when the skin is wet, the resistance is reduced to around 1000 Ω. This is the reason why repairing the electrical connection with the wet skin is always dangerous.

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